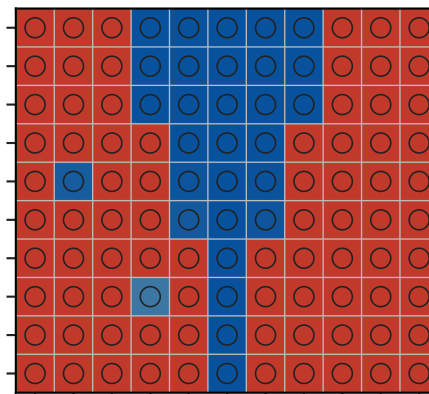
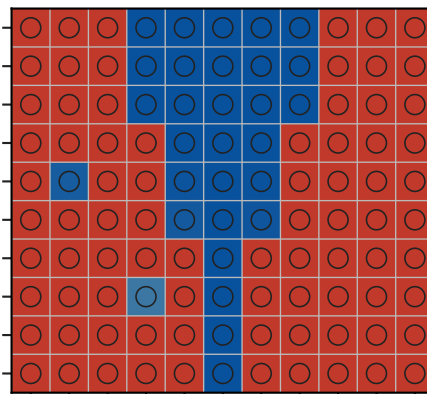
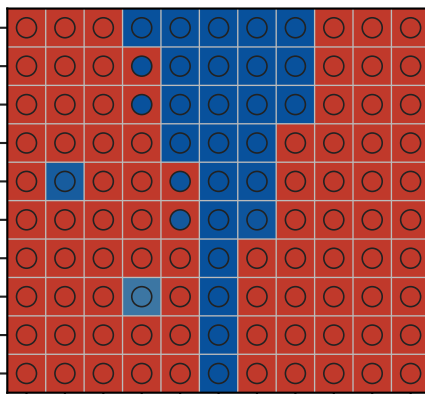
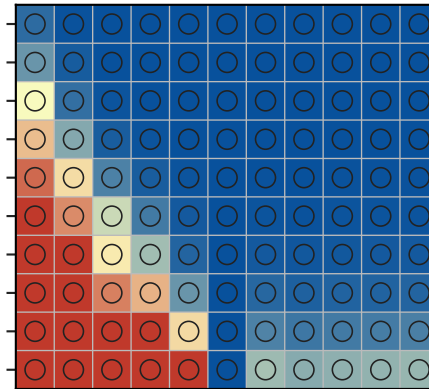
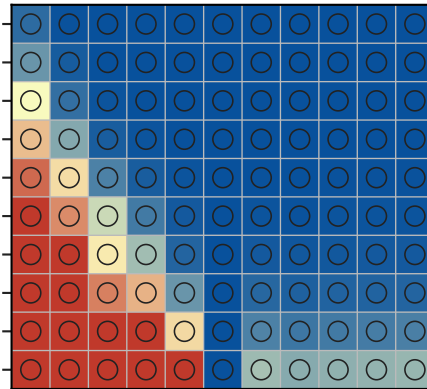
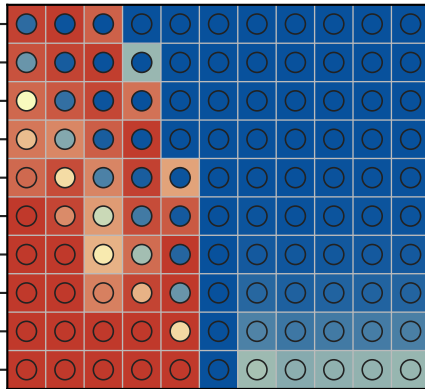


N = 5**N = 10****N = 20**

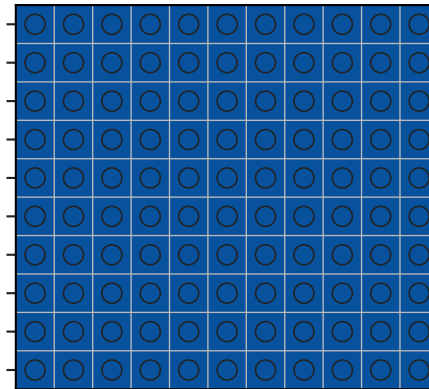
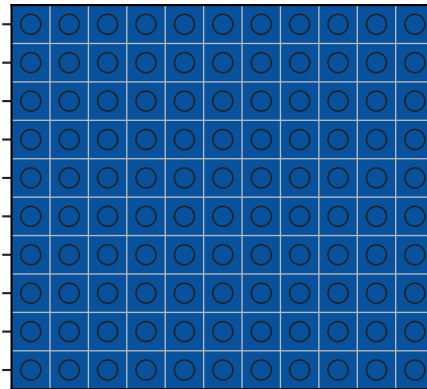
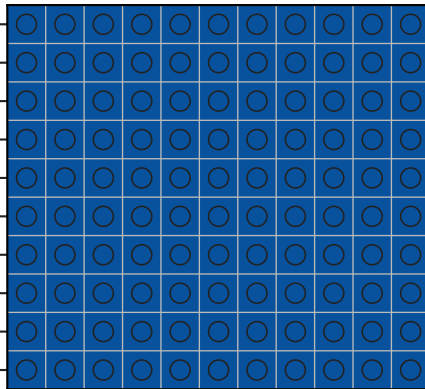
lflank extent

10
9
8
7
6
5
4
3
2
1**BWA-MEM**

lflank extent

10
9
8
7
6
5
4
3
2
1**BWA-MEM
(no-clip)**

lflank extent

10
9
8
7
6
5
4
3
2
1**NW-flex** Δ (Hap N - Ref N) Δ (Hap N - Ref N) Δ (Hap N - Ref N)